

Heredity before individuality in a digital primordial soup

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Theories of the origin of life disagree about what came first but agree about what the transition was: a spreading pattern became an individual, whose future depends on itself rather than on its surroundings^{1–4}. No experiment has let that step be watched and taken apart. We did so in thousands of worlds of random machine code executing one another. Counting fixes the order of events: a copier that writes fewer symbols than it reads cannot finish a copy of itself in one pass, so it must run on into its surroundings or come back on itself. The first replicator takes the first way: a two-byte word that writes itself when executed, whose offspring are an exact copy in only half of its encounters. Evolution's first act is the second way: one jump that returns execution inside the organism, after which the offspring are one string in every context. Individuality so measured arrives as a return in control flow, not as a boundary in space. In a second machine one instruction decides whether life begins open, and the lethality of the sterile order decides how the open phase ends. The first living things are the simplest objects their chemistry makes.

Main

What distinguishes a living thing from the chemistry around it has been answered in two ways. One tradition lists properties, metabolism, compartment and heredity, and organises the origin of life around which came first^{8–11}. The other locates life in a relation: a living system is one whose operations produce the organisation that produces them, so that its future depends on its own past rather than on its environment, in Maturana and Varela's operational closure¹, Rosen's closure to efficient causation², Fontana and Buss's organisational closure³ and, made quantitative, the information-theoretic individuality of Krakauer and colleagues⁴. This view coincides with the moment population genetics identifies as the start of Darwinian evolution proper, when vertical descent comes to dominate the communal exchange of an early world^{14,17}. Whatever the first replicating molecules were, this is the transition that made them organisms, and no experimental system has yet allowed it to be observed as it happens, repeated under controlled variation, and taken apart.

Soups of random programs come close. When random byte strings for a simple machine are made to execute one another, with no fitness function and no replacement, self-replicating programs arise and take over⁵, as they did earlier in the Coreworld⁶ and among random Lisp expressions⁷. For the Z80 processor the first replicators are known to be stack-based load–push programs that give way to a block-copy family after a period of zero-poisoning^{5,12}, the takeover has been attributed to mutational robustness¹², and self-replication has been detected by serial transfer into fresh random tapes³⁷. What has not been measured is what separates the first replicator from its successor besides robustness to mutation: how far each depends on the program it happens to meet, how that dependence is lost, which instructions of the machine decide it, and what order of events a machine without a literal-writing instruction produces. We treated such a soup as a test of one explanation. A copier that writes fewer symbols than it reads cannot complete a copy of itself in one pass over its own cells, so its first pass must end either by running on into whatever lies beyond it or by returning to a cell it has already executed. If life in a soup begins with the smallest thing that writes itself, it must begin open, and individuality must be a later invention that closes the path of execution. We ran 3,560 worlds of random Z80 code across thirteen deletions of instruction families, program lengths from 3 to 100 bytes, execution budgets, mutation rates and memory geometries, with confirmatory stages pre-registered with kill criteria (Methods). Heredity is measured by intervention: a program is lifted out, executed against fresh random partners, and its offspring are tested in turn, so that a pattern that merely spreads is not mistaken for one that inherits; the information an offspring carries about its partner is measured the same way. We then repeated the central experiment on a second, minimal machine whose instruction set we could edit, and proved what could be proved.

Sterile order precedes life

The soup holds 20,000 programs of L bytes on a 160×125 lattice (Fig. 1a). At each step 8,192 random cells are each paired with a lattice neighbour, the pair is executed as one program for 128 instructions and written back, and a background mutation replaces a fixed number of random bytes; nothing else happens (Methods). A tape is *heritable* if, lifted out and

executed against random partners, it makes copies that copy in turn; emergence is dated by the first heritable tape among the most common, and the heritable fraction of random cells is the population measure.

In every unablated world the first thing to form is order that does not inherit. At the moment the most common pattern first occupies a tenth of the soup, between 24% and 34% of all bytes are zero at every tape length from 9 to 100 (Fig. 2). The zeros are self-written: push, call and reset instructions executed with empty registers write zeros wherever the stack pointer points, and zero is itself an instruction that does nothing, so the pattern spreads by being written and changes no state when executed; its carriers produce no copies that copy in turn. We call this flood *tar*, after the asphalt that prebiotic chemistry makes from the reactions that also make sugars¹³: order without heredity, made by the chemistry that will make life.

The tar is what detectors see. The high-order entropy used to detect the transition in random-program soups⁵ is anti-correlated with life in our worlds of 25 to 81 bytes (area under the curve 0.07–0.34 against the culture-test ground truth), because the flood compresses better than the replicators that replace it (Extended Data Fig. 5). A complexity-times-abundance measure of selection^{19,43} behaves the same way: it rises a median nineteen-fold above its random-soup value before the first heritable replicator exists in 71 of 80 worlds, and as a detector of heredity across tape lengths it is no better than compression (AUC 0.72 against 0.75; Extended Data Fig. 12). The reason is that the first living thing is not more complex than the tar: the assembly index of the first replicator is the lowest any string of its length can have, equal to that of the modal sterile tape in 79 of 80 worlds. Order comes first, it is as simple as what follows, and only intervention tells the two apart.

The first replicator is a literal

The first heritable replicator is the same program almost everywhere: the two-byte word 01 c5, repeated to fill the tape (Fig. 1a). In the pre-registered confirmatory stage it is the modal first tape in 19 of 20, 12 of 20, 14 of 20 and 18 of 20 worlds at L = 16, 20, 50 and 64, the rest being its register variants, at a Kaplan–Meier median of 450–750 steps whatever the length (Fig. 1b). Executed, the word is an instruction that loads the next two bytes into a register pair, followed by an instruction that writes that pair into memory at the stack pointer, which starts in the partner. The bytes the first instruction carries are the bytes the second writes: the program is its own data, and it copies itself as a side effect of being run. It has no loop, because the ring returns the pointer; no counter, because every byte of the tape is the same word; and no test of its surroundings, which it executes but never inspects before writing.

It is also nearly the only thing of its size that can replicate. Of all 65,536 two-byte words tiled to 16 bytes, exactly five copy themselves into an all-zero partner, all of them load-and-push pairs, and none copies into more than 81% of random partners (Extended Data Fig. 8). A further 246 words pass the heredity threshold without producing a single faithful copy, by smearing return addresses, which is why faithfulness as well as heredity is required before anything is called a replicator.

Openness limits the first replicator

The first replicator is *open*. Executed against 256 random partners it copies into a median 0.66–0.73 of them (range 0.56–0.84 over 80 worlds and four lengths) and loses a quarter or more of its own bytes in a median 0.06–0.43 of encounters (Fig. 3a,b). The failures have one cause: after writing its copy the pointer runs on into the partner and executes the partner's bytes, which can poison the register pair before the next push. Against all-zero partners the pusher copies perfectly; its success falls as the density of active bytes in the partner rises, and no execution budget rescues it.

Openness has a measurable meaning. With the organism fixed and execution deterministic, the information that flows from the partner into the offspring equals the entropy of the offspring over random partners (Methods). For the first replicator the offspring is an exact copy of the parent in a median 56% of encounters at L = 16, 29% at L = 20, 5% at L = 50 and 0.8% at L = 64; every one of its L positions varies across partners in all 80 worlds; and its entropy over 256 partners is a median 7.0 bits, from 3.9 bits at L = 16 to the 8-bit ceiling of the sample at L = 64 (Fig. 3c): what a child looks like depends in part on the neighbour it was made against. The population shows it. At L = 16 the heritable fraction of random cells stays between 0.07 and 0.19 for the first 20,000 steps after emergence (Fig. 3d): the first replicator's reign is a world of damaged copies.

Closure evolves as a control cycle

The first replicator does not stay. By 300,000 steps the dominant tape at $L = 16$ carries a control-flow instruction in 20 of 20 worlds, and 17 of them end on the identical eight-byte sequence whose conditional return sends the pointer back into the organism's own body (Extended Data Fig. 2); at $L = 50$ all 20 worlds carry a relative jump inserted into the pusher tiling, 14 of them byte-identical (Fig. 3e). Across the 233 unablated worlds of the exploratory stages, 92 gained a control-flow instruction between the first replicator and the final dominant and 3 lost one (exact McNemar $P = 7 \times 10^{-24}$). Every such successor copies into 1.00 of 256 random partners with 0.00 self-damage, whichever instruction implements the cycle, a return, a jump or a hardware block-copy loop (Fig. 3a,b,f). Its offspring carry nothing detectable from the partner: in 63 of the 67 worlds whose final dominant has such an instruction the offspring is one and the same string in all 256 encounters (one-sided 95% bound on the rate of variants, 1.2%), and the four exceptions copy all but one to three bytes, which remain the partner's (Fig. 3c). The heritable fraction of random cells rises to 0.75–0.94 when the closed successor takes over (Fig. 3d).

We call this property *closure*: the organism's bytes are unchanged by, and its offspring independent of, every context tested (Methods). Three properties travel together in these worlds and must be kept apart: a loop in the code, which is syntax; confinement of the pointer to the organism's own cells, which is the mechanism; and independence of the offspring from the partner, which is the outcome. All three are measured. On a traced copy of the executor every first replicator fetches partner bytes in all 256 encounters and every loop-bearing final dominant in none (89 of 89 across the confirmatory and control stages); confinement and zero inflow agree in 106 of 110 finals, the four exceptions keeping one to three partner bytes, and no confined final has inflow (Methods). The second machine shows that the three can come apart. In this operational sense closure is operational closure¹ made countable and one component of individuality⁴ made measurable by intervention. It is not a membrane, since the closed organism's bytes are as exposed to writes as before; not a test of the environment, since the return tests a flag the organism set itself; and not a property of place, since a pre-registered control with well-mixed pairing (ten worlds, $L = 16$) found the open phase exactly as heritable as on the lattice and closure later rather than sooner, reached in 7 of 10 worlds against 20 of 20, although the first replicator met its own kind in 57% of encounters on the lattice and 11% when mixed (Methods). It is a cycle in the organism's control flow that never leaves the organism (Fig. 3f). The cycle is made in two ways, and one of them depends on a convention. Pair memory is addressed modulo $2L$ while the program counter is 16-bit, so a backward relative jump that crosses address 0 lands at the true-ring target only when $2L$ divides 65,536 ($L = 16, 32, 64$); at $L = 20$ and 50 it lands elsewhere inside the organism, and all 20 closers at $L = 50$ and 5 of 19 at $L = 20$ use such a jump (Methods). The other 42 closers, returns and block copies, need no such help. At aligned lengths, where no closer can use the wrap, closure is slower and monotone in L : 20 of 20 worlds by 300,000 steps at $L = 16$, 12 of 20 by one million at $L = 32$ (a pre-registered test added in revision, all twelve by block copy) and 8 of 20 at $L = 64$, where the open pusher is also the least damaged open form in the atlas (self-damage 0.06) and so under the weakest pressure to close; the pre-registered expectation of at least 10 of 20 closures at $L = 20$ and 64 was met at $L = 20$ only by a family our syntactic criterion did not count, and not at $L = 64$, and the expectation of 15 of 20 at $L = 32$ was not met (Methods).

Closure trades variation for fidelity. If closure made the organism an individual in the full sense, it should also have made it a better carrier of inherited differences; a pre-registered single-byte mutational scan of all 160 first and final tapes says the opposite (Extended Data Fig. 14). The first replicator is at least as robust to mutation as its successor at every length (a median 0.74 against 0.59 of single mutants remain heritable at $L = 16$, 0.99 against 0.95 at $L = 50$), so robustness to mutation, the explanation offered for the same succession elsewhere¹², does not account for it. And the first replicator transmits its mutations: a changed operand byte is pushed into the copy and pushed again by the copy, at a median 23 of 50 and 32 of 64 positions. The closers transmit almost none, because they execute every byte they copy, so a change is either lethal or corrected: a median 1 of 50 positions for the pusher with its jump, 0 for the return-based and block-copy designs. The two forms of heredity differ in where variation lives: the pusher's transmissible positions are all bytes it executes, operands read as instruction stream and pushed as data, whereas the closers' few transmissible positions are bytes they copy but never execute (82 of 518 sites among the final dominants, all in the five worlds whose genome holds a stretch that a block copy transmits and a jump skips). Variation inherited through bytes that are never run is a genotype in Langton's sense²², copied as data and not read as code, and here it is a late and rare acquisition. The first closed organisms are one string in every context and one string across their own mutants.

The return is what selection acts on. Seeded at 1% into a world filled with its own ancestor, the pusher with its jump holds half the cells within 110–130 steps in 5 of 5 soups, about forty encounters per cell, and the ancestor seeded into a world of the closed form leaves no trace; in pusher-filled worlds with no seeding the jump arises by mutation and sweeps within 5,000–13,250 steps in 3 of 3, where a random start takes of order 10^5 (Methods). The advantage measured in isolation, 1.00 against 0.69 of partners copied, is a competitive advantage among kin, and it is paid for with the variation channel.

Three instruction families are load-bearing

Which instructions does this need? The seed-paired ablation atlas (Fig. 4a) finds three load-bearing parts and nothing else. Removing *immediate loads* leaves 2 of 10 worlds alive at 300,000 steps; removing *stack writers* delays emergence 160- to 992-fold in median and hands the soup to a replicator built on the block-copy instruction; removing *all copying* (112 opcodes) or leaving only read-modify-write instructions produces no replicator in 40 worlds at one million steps. Every other ablation, block copy, calls and returns, memory loads, input and output, exchanges, is within seed-paired noise of the unablated soup.

The size axis has no floor but a dead zone (Fig. 4b): emergence is certain from $L = 16$ to 100 with a median time rising only eightfold, because tiling makes a large organism of a two-byte word, but only 4, 4 and 6 of 10 worlds live at $L = 9, 10$ and 12, where the descending stack pointer and the ascending instruction pointer collide on the short ring and poison the register pair; padding the ring without changing the organism switches the zone off (Fig. 4c; Extended Data Fig. 3). Tar can prevent life as well as precede it: at $L = 9$ removing the stack writers makes life *more* likely (12 of 20 against 6 of 20 worlds at 128 steps, 19 of 20 against 5 of 20 at 512; Cochran–Mantel–Haenszel $P = 3.9 \times 10^{-6}$; Fig. 4d), because their tar destroys nascent block-copy replicators before they assemble, which reach the ten most common tapes in 5 of 20 worlds at a median 118,500 steps with stack writers present and in 20 of 20 within 3,400 steps without. The execution budget decides whether an organism can be faithful: at $L = 100$ the block-copy regime is faithful in 1 of 10 worlds at 128 steps per encounter and 10 of 10 at 256, while at 64 steps no exemplar ever reaches 0.5% of the soup yet 6 of 10 final soups are heritable populations with no clone, heredity carried by partial copies, the communal regime placed before the Darwinian threshold¹⁴ (Extended Data Fig. 4).

One instruction decides how life begins

If the order of events is forced by the literal channel, a machine without one should be born closed, and giving it one should start life open. BFF, the Brainfuck family of Agüera y Arcas et al.⁵, has no literal channel: its instructions increment in place or copy from memory, and no execution of fewer than 129 instructions can copy a 64-byte tape with two or more distinct bytes (Methods); a search over 16.4 million straight-line programs found only one-symbol fills, the BFF analogue of tar. Eighty-four soups of 2¹⁷ programs run for 16,384 epochs test the prediction (Fig. 5). In BFF as published, 9 of 24 worlds produced a heritable replicator; every first replicator contains a loop (9 of 9) and none is open (0 of 9). A wrapping pointer gives more transitions (19 of 24) but the same birth: 19 of 19 loop-bearing, 0 of 19 open. Adding one instruction that writes its own two operand bytes below the write head changes the beginning completely: in 12 of 12 worlds the first replicator is the one-byte tiling of that instruction, whose operands are two more copies of itself, loop-free, with its pointer entering the partner in every encounter, present at the first sample with 53–60% of the soup. It is the 01 c5 of BFF, and its offspring carry 8 bits of the partner, the ceiling of the measurement (Extended Data Fig. 11).

Then it dies: within about 200 epochs each soup turns into tar that halts a pointer within a few dozen steps, since an unmatched bracket ends BFF execution; copies into such partners are cut short and sterile, and the open replicator persists only as a 0.2% minority, heritable in isolation and unable to spread; no closed class evolves, and none was found among 1.3 million periodic programs of this language up to period 10 (Methods). The open phase is a *window*, and one more switch sets its length: making an unmatched bracket a no-op, benign tar, leaves the same open replicator in place in 12 of 12 worlds for all 16,384 epochs with every random tape heritable at the end (Fig. 5c), a substrate in which life stays open for as long as we watched. Open here means that execution enters the partner; by the information criterion this organism is already closed, because under a wrapping pointer its pushes lap the ring and overwrite everything, so its offspring is the same string in nearly every context (0.04 bits) and it never needs a cycle. The first machine completes the lethal column. Making a zero byte halt execution, so that the Z80's own tar becomes lethal, removes the open beginning altogether: in 10 of 10 pre-registered worlds no literal pusher ever establishes, and life begins closed and late, as a block-copy replicator with a hardware loop that contains no zero byte and copies 1.00 of partners with 0.00 self-damage from its first appearance, at a median of 46,250 steps against 525 under benign tar (Methods; Extended Data Fig. 13). A second real instruction set tests how far this reaches. Restricting the Z80 to the Intel 8080's instruction set, by making every Z80-only instruction a no-op (the bit and rotate page, every block copy and extended instruction, the relative jumps and the register exchanges; Methods), keeps the literal word and removes every closer but the return. Three predictions were registered: the load–push word first at both lengths; the return design closing most worlds at $L = 16$; few closures at $L = 32$, where there is no block copy and the return's popped addresses fall in the partner. All three were met: the load–push word first in 40 of 40 worlds (median 650 and 325 steps); at $L = 16$ the byte-identical return design `ad e3 21 e3 21 c0 ad c0`, pure 8080 code and the Z80's own modal closer, dominates 20 of 20 worlds by 300,000 steps; at $L = 32$ none of 20 worlds closes by one million steps and the open pusher reigns throughout. The beginning is the same in both machines; what follows depends

on whether the instruction set can express a closer at that ring size, which is also why long 8080 tapes never produced a looping variant in ref. 5. The fourth cell, no literal channel with harmless brackets, was run after review: 7 of 12 soups produced a heritable replicator, every first replicator carried a loop (7 of 7) and five of the seven kept the pointer inside the organism, but one entered the partner in every encounter, and in four soups the replicator was gone again by 16,384 epochs (Methods). Together these fill a two-by-two map (Fig. 5d): a literal channel decides whether life starts open; the lethality of the tar decides whether an open beginning lasts, or happens at all; and the supply of closed designs decides whether what follows is closure, a late closed beginning, or extinction.

Closure requires a cycle

Two counting statements make the order of events a theorem rather than an observation (Methods; Supplementary Information). A straight-line program's pointer runs on into its partner, so a straight-line replicator is open by construction; and a closed replicator, whose pointer never leaves its own cells, must execute some address twice, because one pass over its cells writes fewer bytes than a copy needs: closure requires a cycle (Fig. 6). A straight-line replicator exists wherever the instruction set can write a literal, content independent of the memory being overwritten, at a self-advancing pointer, and it is the size of the literal; exhaustive execution of every two-byte word shows that in this machine no closed design is as small (Supplementary Information, Proposition 3), so the open form is present in a random soup with exponentially higher probability than any closed one. Without a literal channel the count goes the other way: in BFF no execution of fewer than 129 instructions can copy a tape, so every replicator of that machine carries a loop (Theorem 1). The boundary of the theorem is itself a prediction, and the second machine confirmed it: a copier that writes at least as much as it reads may close by saturation without a cycle, which is what the wrapping literal pusher of BFF does.

In information terms closure is zero inflow from context to offspring, $I(o; e | x) = 0$, the limit that perfect fidelity requires (Methods, Lemma 7); perfect fidelity requires zero inflow, and Fig. 3c shows the measured inflow reaching zero as fidelity reaches one. What the theorems do not decide is the dynamics after the open beginning. Closure arrives with probability about $1 - \exp(-q \int n \, dt)$, where $\int n \, dt$ is the integral of the open population over time and q the rate of closing mutations; benign tar lets the integral grow until a closer is found, lethal tar closes the window first, and whether a copyable closer exists decides what a lasting window can become (Extended Data Fig. 11).

Discussion

The order of events is not a contingency of one machine. Given a machine's default flow and copy cost, it follows from counting: whatever writes fewer symbols than it reads must, on its first pass over itself, either leave itself or return, and the smallest thing that writes itself has no room for a return. So the first replicator is open, its offspring an exact copy in only half of its encounters; closure is the return, after which the offspring is one string in every context, reached by four different instructions in different worlds. The explanation has few adjustable parts: the dichotomy is arithmetic once the machine's default flow and copy cost are given, the literal channel is a property one reads off a substrate, and the lethality of by-products is a measurement. It reaches beyond its origin, to any sequential copying process with overhead, and it answers questions it was not built for: why the first replicators are sloppy and only partly heritable; why individuality can precede compartments; why a cycle, not a membrane, is the first thing selection builds; why complexity-based signatures of life can fail at the origin, the first living thing being no more complex than the sterile order that precedes it; and why the first innovations after replication should concern control, where copying starts, stops and returns, rather than function or code. The boundary of the rule is itself a prediction, and the second machine confirmed it: a copier that writes at least as much as it reads can close by saturation, overwriting everything it touches, and needs no return; the literal pusher of BFF under a wrapping pointer does exactly this, entering its partner in every encounter yet importing nothing from it. And when the literal road is blocked, by removing the stack writers, by a language without a literal channel, or by tar that halts whatever executes it, life still comes, but late and born closed: the theorem's exponentially rarer closed beginning, visible as a ninety-fold delay.

These are predictions for chemistry, not results about it, and they can be tested. Template copying is a literal channel: a strand specifies its complement by physical fit, with no interpreter, so the first replicators in such a chemistry should be the shortest self-templating repeats, dependent on their surroundings and productive mainly of partial copies; the experiment is to measure heritable propagation among sloppy copies by transfer into fresh material, and openness as the variation of the product with its context. By-products are a variable, not a nuisance: whether a chemistry's tar is inert or inhibitory to the copying process should decide whether an open phase lasts long enough for control to evolve, which a screen of nonenzymatic systems with matched inert and inhibitory additives can test¹³. Processivity is the chemical write

ratio: vertical lineages should appear when one copying event spans the genome, as polymerase ribozymes of increasing processivity could test⁴⁴, and in laboratory evolution of RNA replicators the earliest fixed changes should concern control, circularisation, terminators and primers⁴². For the search for life elsewhere the lesson is methodological: abundant low-complexity order precedes and surrounds the first replicators and trips compression and assembly measures alike^{19,43}, so detection should be by intervention, adding fresh substrate and looking for propagation that depends on the inoculum, and order without heredity should be sought as a precursor signature. What limits the claims is plain: one real instruction set, its 8080 subset, and one toy language; units of fixed size, start address and pairing imposed by the harness, within which autonomy is studied; a culture test that is our own ground truth, though the order of events held under every setting we tried; closure at the largest tape lengths in progress rather than complete within our horizons; and local interactions, though a pre-registered well-mixed control found the order unchanged and closure later rather than sooner⁴¹. Within those limits the result stands: heredity came first; independence from context came later, as a return in control flow; and the step that made a replicator an organism, in the operational sense measured here, was the step that made its offspring stop depending on its environment.

Methods

Soup. The soup holds $N = 20,000$ tapes of L bytes (L from 3 to 100), initialised with uniformly random bytes, on a 160×125 square lattice with local interaction. At every step 8,192 cells are drawn uniformly at random and each is paired with one of its four lattice neighbours (a step off the edge is reflected; the lattice does not wrap); a pair is active if neither member is already in another pair that step, which leaves $\approx 3,980$ active pairs (0.395 active interactions per cell per step; the count varies by a few per cent with concurrent GPU load and does not affect results). An active pair is concatenated, organism first, into a pair memory of P bytes ($P = 2L$ except in the padding experiment of Stage F, where $P \geq 2L$ and the padding bytes are never written back), executed by the Zilion Z80 core (@neovand/zilion 0.2.0, WGSL) for a fixed budget of 128 instructions (32, 64, 256, 512, 1,024 or 2,048 in the budget arms) from byte 0 with all registers zero and the stack pointer at the last byte of the partner, and both tapes are written back. After each step $8,192/2^k$ bytes chosen uniformly at random over the soup are replaced by random bytes ($k = 4$ unless stated; $k = 1-6$ in the mutation arms). There is no fitness function, no replacement and no death. Instruction-set ablations replace whole opcode families by no-ops through a prefix-aware fetch hook that sees the operation after DD/FD and ED/CB prefix resolution; the thirteen ablations and their opcode counts are listed in Extended Data Table 2 (none, block-copy (4 opcodes), stack-writes (46), stack-write-only (22), stack-read-only (24), push-only, ex-sp-only, call-rst, call-rst-write, ld-mem, ld-imm, ld-reg, all-ld, ed-loads, cb-page, no-copy (112), rmw-only). The emulator was validated against an independent Z80 reference with zero genuine divergences in randomised 5,000-pair tests with and without ablation; it is not treated as ground truth for the Z80 itself, and no claim depends on cycle-exact timing.

Recording. Every run writes a sample at fixed steps (every step for the first 30, then a geometric schedule, then every 500 or 1,000) with: the ten most common exact tapes and their shares; quasispecies occupancy (fraction of cells within Hamming distance $\lceil L/4 \rceil$ of the dominant tape); byte histogram, zero-byte fraction and byte entropy; brotli-compressed bits per byte (quality 2) and the high-order entropy of ref. 5; species entropy and unique-tape fraction; per-interaction statistics (bytes written into the partner, copy events, destroyed programs); 16 random tapes for the functional census; and full soup snapshots at fixed steps and at emergence.

Culture test and partner tests. A tape runs as the first program against 64 uniformly random partners for the condition's budget; copy success is the fraction of partners that become $\geq 75\%$ copies at the best cyclic shift; gen2 is the copy success of the produced copies against fresh partners, weighted by the copy fraction; heritable means $\text{gen2} \geq 0.3$, faithful additionally $\geq 50\%$ of partners copied. Partner-independence tests use 256 random partners and record self-damage ($\geq 25\%$ of the organism's bytes changed in one encounter). In BFF the culture test also records whether the instruction pointer entered the partner in each encounter; closed means $\leq 5\%$ of encounters entered and $\geq 95\%$ of partners copied, open means $\geq 50\%$ entered. Information inflow from the environment, $I(o; e | x) = H(o | x)$ for a deterministic executor, is the plug-in entropy in bits of the offspring (the partner's bytes after the encounter) over 256 random partners, ceiling 8 bits.

Events and statistics. t_{rep} is the first sampled step at which one of the three most common tapes has a share $\geq 0.5\%$ and is heritable by the culture test (a clonality criterion by design: it dates the first *clone*, not the first heritable material); t_{faith} additionally requires faithfulness; $t_{q_{10}}$ (quasispecies occupancy $\geq 10\%$) is the pre-registered occupancy event and is reported because it fires on zero floods as well as on replicators. Emergence fractions are quoted as worlds reaching the event within the horizon with Wilson 95% intervals; times as Kaplan–Meier medians with censoring at the horizon. Ablation effects are ratios of Kaplan–Meier medians with seed-paired sign tests (the same seeds run under every ablation); confirmatory contrasts use one-sided Fisher exact tests and the Cochran–Mantel–Haenszel test pooled over budgets; the

paired gain of control flow uses the exact McNemar test; between-variant transition rates in BFF use two-sided Fisher tests and Mann–Whitney tests on event times with non-transitioning runs censored at the horizon. In BFF, where replicator populations are quasispecies with small exact-class shares, the “first replicator” is the most common tape class (a tape and its reverse count as one class) at the first sample at which it is heritable, and the population event is a heritable fraction ≥ 0.5 of 32 random tapes; t_{rep} is reported alongside. All thresholds (0.75 copy, $gen2 \geq 0.3$, 0.5% share, 10% occupancy, 95%/5% closure) were fixed before the confirmatory stages. Same-seed Z80 runs diverge because GPU scheduling makes the pairing order non-deterministic, so every Z80 claim is a population claim over seeds (ten same-seed repeats spread as widely as ten seeds); the BFF soup is bitwise reproducible across machines (identical epoch statistics on an Apple GPU and an NVIDIA L40S for six seeds over 16,384 epochs).

Definitions and theorems. A *self-writer* is a tape that, executed against the all-zero partner, leaves the partner holding a copy of itself. *Closure*: the organism’s bytes are forward-invariant under execution for every context in the class of uniformly random partners; equivalently its pointer never executes a partner address. Theorem 1: any execution that copies a 64-byte BFF tape runs at least 129 instructions, so every BFF replicator executes some address twice, that is, carries a loop. Theorem 2: in any machine whose pointer advances by default, a closed full replicator executes some address twice, provided a copy costs more instruction executions than the organism has addresses (BFF) or execution cannot stop inside the organism before the budget is spent (the Z80 soups, budget $128 > L$). Theorem 6: a self-consistent literal-write instruction gives a straight-line, hence open, replicator the size of the literal; without one, under the copy-cost condition, every replicator contains a cycle. Lemma 7: $I(o; e | x) > 0$ implies imperfect fidelity across contexts; perfect fidelity implies $I = 0$. Proposition 3 (computer-assisted): the two-byte words fixed under their own execution against the zero partner are exactly the five load–push words, all straight-line. Full statements and proofs are in the Supplementary Information.

BFF. cubff default semantics verified from source: two 64-byte tapes concatenated, pointer and heads at byte 0, ten instructions, every other byte a no-op, unmatched brackets halt, 2^{13} steps per encounter, 2^{17} programs, random ordered pairs each epoch, mutation 2^{-12} per byte per epoch. Switches: `ip_wrap` (pointer wraps modulo 128); `literal` (byte 0x50 = P a b: write a at head1–1 and b at head1–2, then head1 $\leftarrow$ 2, skip the operands); `nohalt` (unmatched brackets are no-ops). Our GPU kernel reproduces cubff’s validation replicator and matches a CPU reference on randomised tests. 24 seeds each for the published and wrapping variants, 12 for each literal variant, and 12 for the published rule with no-op brackets (run after review, pre-registered: every first replicator loop-bearing, met 7 of 7; a transition in at least 9 of 12, not met at 7; pointer-closed first replicators in every world, not met in one); events every 64 epochs; high-order entropy with brotli quality 2 as in ref. 5.

Searches. Two-byte census: all 65,536 words tiled to 16 bytes, culture test (32 partners) and partner test (256), plus the deterministic fixed-point test against the zero partner. BFF straight-line search: 16,447,860 programs (prefix ≤ 3 instructions + tiling unit of period 2–5 with a copy instruction), 2 random partners, wrapping pointer, alphabet prefilter then exact similarity, full culture test for survivors. Closed-replicator search: all 1,309,528 periodic tilings of {P, [,], no-op} up to period 10 under the literal, wrapping, no-halt rules, three partners each.

Mutational scan. For each Stage G first replicator and final dominant, every single-byte mutant (all 255 alternative values per position at $L = 16$ and 20; 32 seeded values per position at $L = 50$ and 64) was run as organism A against 32 shared random partners for 128 instructions and its offspring against 32 fresh partners, with the culture test’s thresholds. A mutant is heritable if $gen2 \geq 0.3$; its mutation is transmitted if the mutant byte is present at the aligned position in at least half of the offspring that are $\geq 75\%$ copies; a position is a transmissible site if at least half of its tested values are both heritable and transmitted; capacity is $\sum_i \log(1 + 255 f_i)$ over positions, f_i being the fraction of transmissible values at position i . Three predictions were written before the run (the successor more robust to mutation than the first replicator; the closed form with more transmissible sites; the matched pusher-with-jump of $L = 50$ with more sites than its pusher); all three failed and are reported as such.

Pair invasion. Soups of 20,000 cells at $L = 50$ under the Stage G dynamics, every cell set to one tape of the matched pair (the pusher tiling, or the same tiling with $20 \neq 0$ inserted three times, the final dominant of 14 Stage G worlds) and 1% of cells to the other at step 0; five seeds per direction, three seeds per resident with no invader; 20,000 steps; the share of cells within Hamming distance 13 of either tape (nearest class), the share carrying the jump word, the zero fraction, and the heritable fraction of 16 random cells every 2,500 steps. The pre-registered outcome, the invader’s class share passing 50%, could not be met at this length because the quasispecies cloud splits between the two classes (a closed tape that has lost two jumps is nearer the pusher); the share of cells carrying the jump word, reported instead, was defined after the data were seen.

Address mapping. Pair memory is a ring of $2L$ bytes: every memory access uses the 16-bit address modulo $2L$, and the stack pointer starts at the largest 16-bit value that aliases the last byte of the partner. The program counter and the stack

pointer themselves are 16-bit, so a relative jump or a stack move that crosses address 0 reaches $(65,536 + \text{target}) \bmod 2L$, which equals the true-ring target only when $2L$ divides 65,536. Of the 67 closed final dominants of Stage G, 25 (all 20 at $L = 50$ and 5 of 19 at $L = 20$) use a relative jump whose target depends on this aliasing; the pair-invasion and mutational-scan results at $L = 50$ are results about the machine so defined. Stage K ($L = 32$, 20 worlds, one million steps, seeds 5001–5020, pre-registered: first replicator a load–push word in ≥ 18 of 20, a closed successor in ≥ 15 of 20, kill below 10) was run after review to test closure at an aligned intermediate length: 20 of 20 and 12 of 20, all closers block copies.

Traced execution. A derived copy of the culture-test executor records every address fetched as instruction stream (opcodes, prefixes, operands) in a 256-bit map per encounter, with byte-identical execution (checked). Pointer confinement is the fraction of 256 encounters with a seeded random partner in which no partner address is fetched; the executed set is the union of fetched organism positions; a transmissible position of the mutational scan is called unexecuted if it is never fetched.

The 8080 subset. The `i8080` suppression makes every Z80-only instruction a no-op: the 256 CB-page opcodes, the 78 defined ED-page opcodes (block copies and their repeats, 16-bit ADC and SBC, IN and OUT (C), NEG, RETN and RETI, IM, LD A,I and A,R, RLD and RRD) and EX AF,AF', DJNZ, the five JR forms and EXX. Two deviations from a real 8080 are kept: IX and IY prefixes still select the indexed form of an 8080 instruction, and suppressed bytes are no-ops rather than the 8080's undocumented aliases. Stage M: Stage G conditions at $L = 16$ (300,000 steps) and $L = 32$ (one million), 20 worlds each, seeds 7001–7020, with the three predictions registered before the run.

Pre-registration. Stages D, F and G and the BFF cells were pre-registered with predictions and kill criteria before launch; every amendment is dated in the change log. Misses reported: the heritable-fraction thresholds of G1(c) failed at $L = 50$; the syntactic loop criterion of G2 omitted hardware loops (6/20 and 3/20 syntactic against 19/20 and 8/20 behavioural); the mechanism first claimed for an open BFF replicator under a wrapping pointer was falsified by the harness test before any soup ran and the prediction restated with both readings; a held prediction for the non-wrapping literal cell was corrected by the counting argument before it was run.

Experimental stages. Stage A (630 runs, $L = 16$): seven ablations $\times$ budgets {32, 128, 512} $\times$ mutation {1/4, 1/16, 1/64}, ten seeds, 300,000 steps. Stage B (640): tape lengths 4–100 under two ablations and three budgets. Stage C (490): thirteen ablations at $L = 16$, seeds 101–110, with one-million-step nulls. Stage D (280, confirmatory): the $L = 9$ reversal with seeds 1001–1020. Stage E (1,230): the size axis under four control arms, budget and mutation sweeps at $L = 100$, variance and exception cells. Stage F (210, confirmatory): padded pair memories. Stage G (80, confirmatory): closure with seeds 2001–2020 at $L = 16$ and 50 (300,000 steps) and $L = 20$ and 64 (1,000,000 steps). BFF: 24 seeds each for the published and wrapping variants, 12 for each of the three literal variants. Stage H (10, control): well-mixed pairing, the partner drawn uniformly from the soup, at $L = 16$, seeds 3001–3010, run locally. Stage I (10, control): a zero byte fetched as an opcode halts the pair for the rest of the encounter (lethal tar), at $L = 16$, seeds 4001–4010, run locally, with the culture and partner tests under the same rule. Post hoc analyses (closure, two-byte census, detectors, invasion, census dynamics, ring sweep) used recorded data and local GPU time only.

Compute, code and data. 3,560 Z80 runs and 84 BFF soups on Modal L40S GPUs plus local runs on an Apple GPU; $\approx$ \$400 at list price. The source code (simulator, assays, analysis and figure scripts), every condition file, every raw run summary, every generated table and the pre-registration documents with their dated change log are in the project repository (to be archived with a DOI on acceptance); each figure is produced from the generated tables by one script, and each number in the text is traceable to a named table.

Figure legends

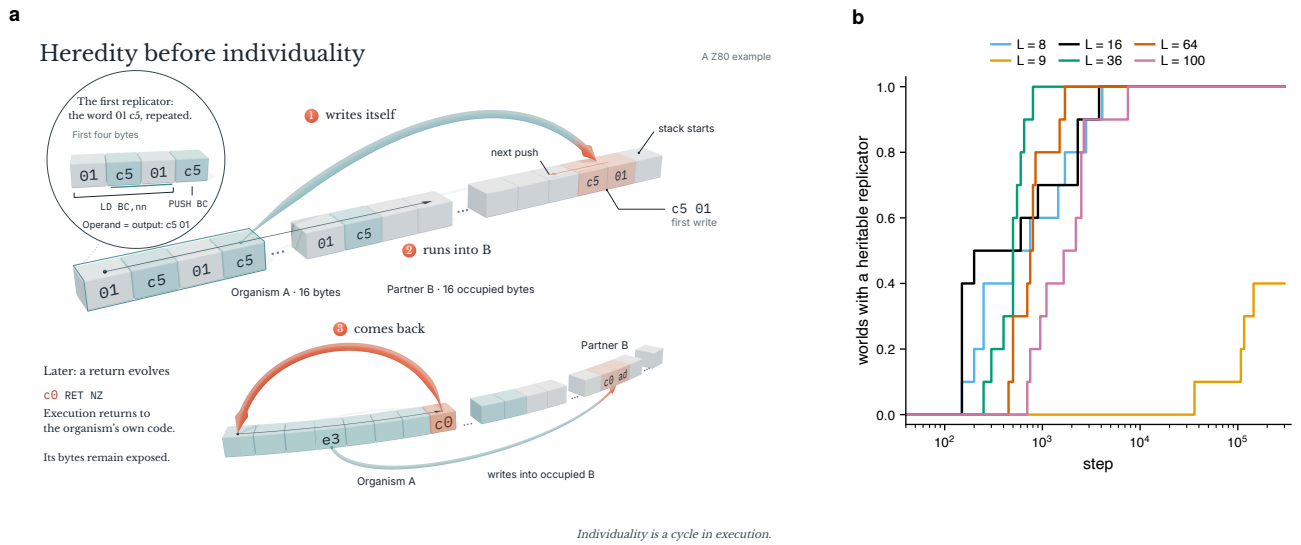


Fig. 1 | The first replicator and its closure. **a**, The first replicator and its encounter (design panel, provisional). Organism A (teal, 16 bytes) and partner B (grey, 16 bytes) share one 32-byte memory; execution starts at A's first byte and moves right, the stack pointer starts at B's last byte and moves two bytes left with every push. The first replicator is the word 01 c5 repeated: LD BC, nn loads its own next two bytes, c5 01, and PUSH BC writes exactly those two bytes at the stack pointer (operand = output), so the partner fills from its far end with the organism's own word (1) while execution runs on into the partner (2). Later a return evolves (3): execution revisits the organism's own code, and its bytes remain as exposed as before. **b**, Kaplan-Meier fraction of worlds with a heritable replicator by step for L = 8–100 (Stage E, 10 worlds per length, 300,000 steps).

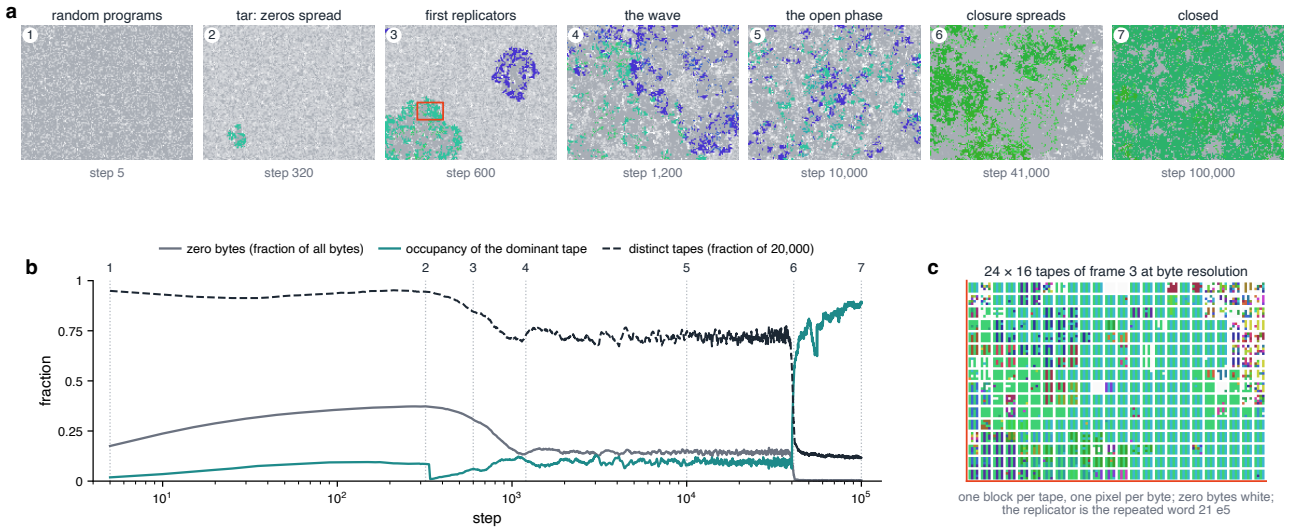


Fig. 2 | One world, watched. Seed 2002 at $L = 16$, run with dense snapshots for Supplementary Video 1. **a**, The lattice (160×125 tapes, one pixel per tape) at seven steps. The four most common tapes with at least 20 copies are coloured by the mean colour of their bytes (the colour code of **c** and of the video); every other tape is grey, lighter the more of its bytes are zero, so the tar shows as bleaching. Zeros spread first (frame 2: 37% of all bytes at step 320); the first replicators appear as patches of their own copies (3; the teal patch is the word 21 e5, the blue one 01 c5, both load-push words); they sweep the lattice (4) and compete for 40,000 steps among pockets of zeros (5); a new class carrying a conditional return (c0, RET NZ) spreads from step 41,000 (6) and takes over while the zeros disappear (7): at step 100,000 the dominant tape occupies 89% of the soup, 2,290 distinct tapes remain of 20,000, and 0.4% of bytes are zero. **b**, The same world's zero-byte fraction, dominant-tape occupancy (cells within Hamming distance 4 of the most common tape) and number of distinct tapes against step; dotted lines, the frames of **a**. **c**, The 24×16 tapes inside the red rectangle of frame 3 at byte resolution (one 4×4 block per tape, one pixel per byte, zero bytes white): the replicator's tapes are the repeated word; the rest is random code and tar.

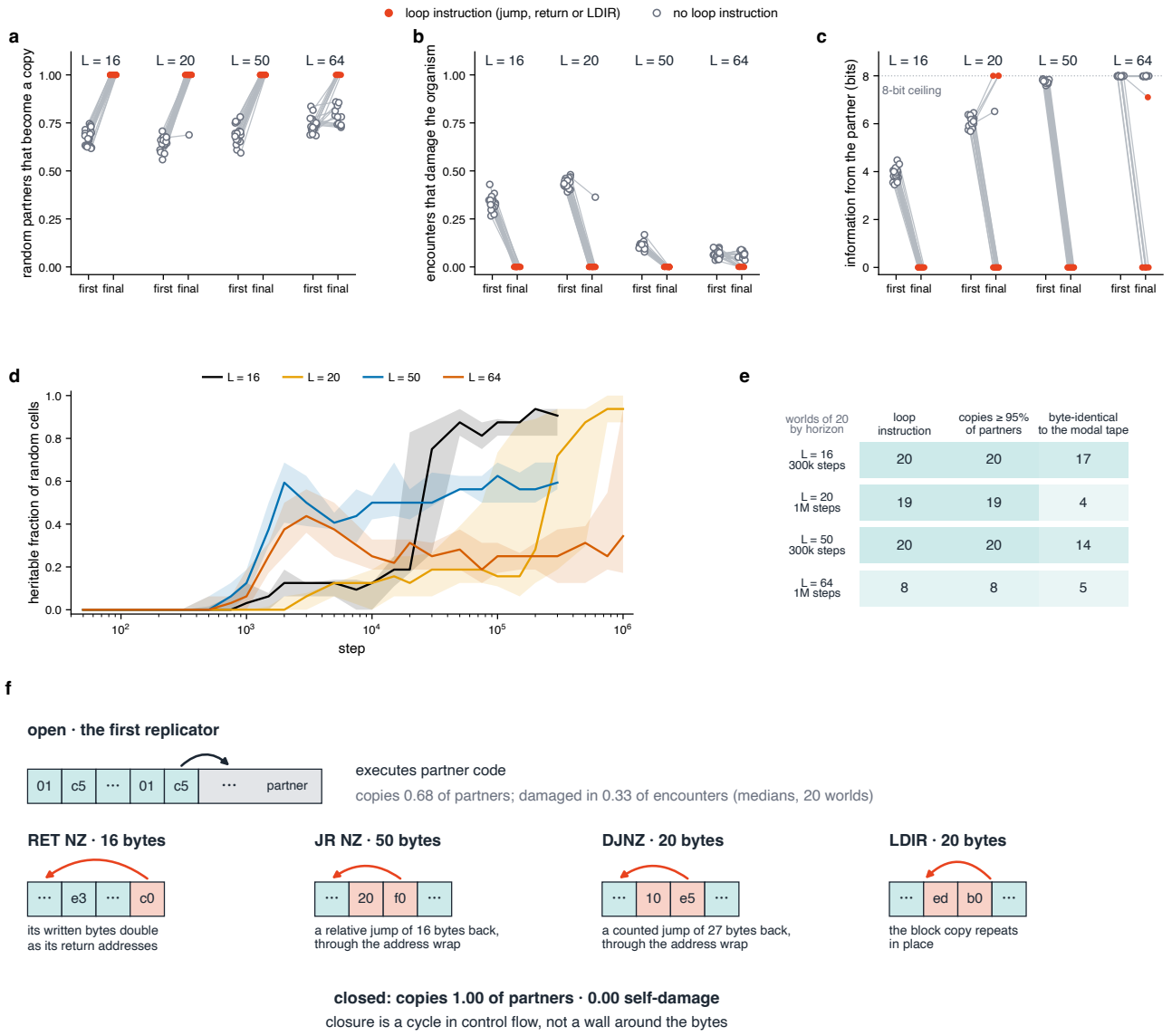


Fig. 3 | The first replicator is open, the successor closed. **a**, Fraction of 256 random partners that become a $\geq 75\%$ copy of the first heritable replicator (first) and of the final dominant (final) of every Stage G world, grouped by tape length (20 worlds per L; grey lines join the same world; open circles, no loop instruction; filled, jump, return or LDIR). **b**, Fraction of encounters in which the organism loses $\geq 25\%$ of its bytes. **c**, Information the offspring carries about the partner, $H(o | x)$ in bits over 256 random partners (dotted line, the 8-bit ceiling of 256 partners). **d**, Heritable fraction of 16 random cells against step (median and interquartile range over worlds). **e**, Worlds (of 20) whose final dominant carries a loop instruction, copies $\geq 95\%$ of partners, and equals the modal final tape byte for byte, by tape length and horizon. **f**, Control flow of the first replicator and of four closed successors, schematically (bytes abbreviated; the instruction that closes the cycle is shaded and its return drawn as a vermilion arc): the pusher's pointer runs on into the partner, whereas RET NZ returns to addresses formed from the organism's own written bytes, JR NZ and DJNZ jump back (through the 16-bit address wrap), and LDIR repeats in place. Medians of the partner test over Stage G worlds: the first replicator copies 0.68 of partners and is damaged in 0.33 of encounters; every closed successor copies 1.00 with 0.00 self-damage. Exact traces, Extended Data Fig. 2.

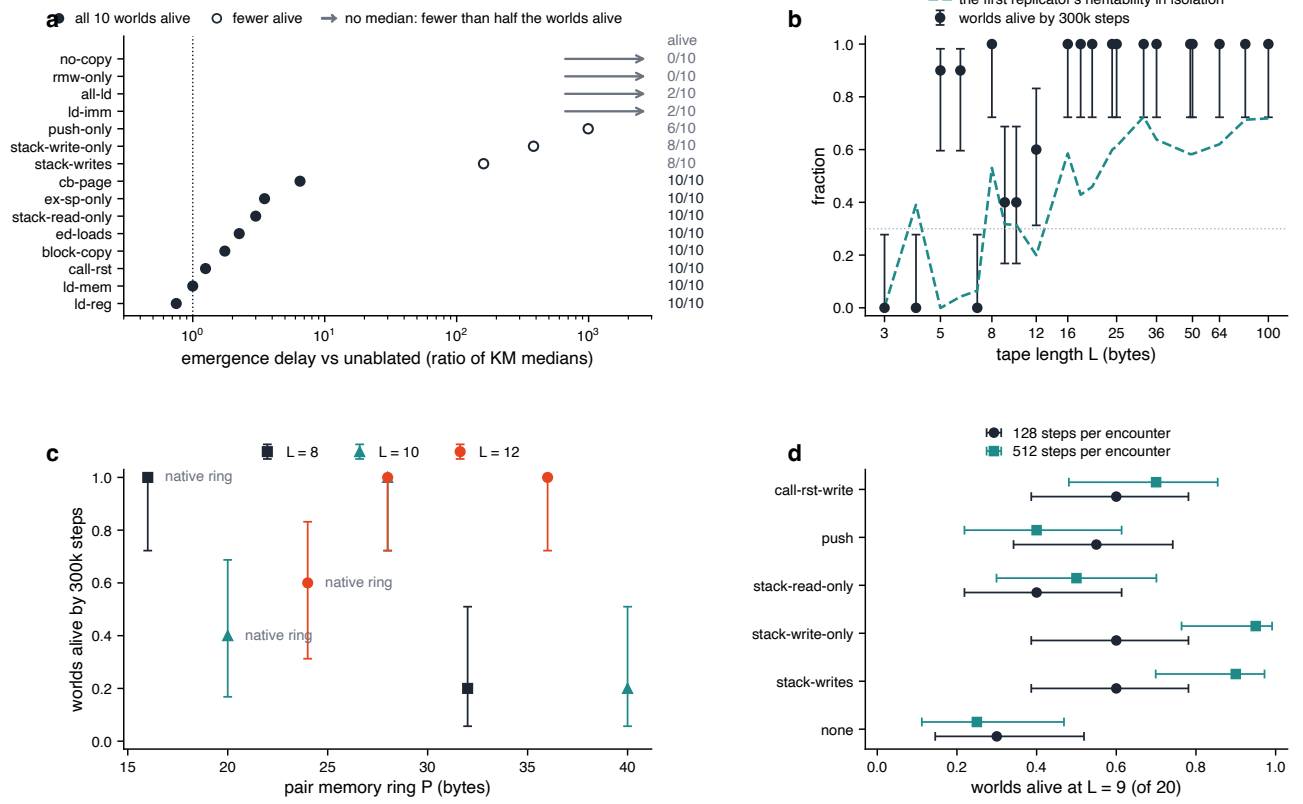


Fig. 4 | What the instruction set must provide. **a**, Delay of emergence relative to the unablated soup (ratio of Kaplan–Meier medians, 128 steps, mutation 1/16, seeds 101–110) for each ablation; filled, all ten worlds alive; open, fewer; arrows, no ratio because fewer than half the worlds produced a replicator; right margin, worlds alive by 300,000 steps. **b**, Worlds alive by 300,000 steps against tape length (Wilson 95% intervals) and the first replicator’s heritability in isolation (dashed; dotted line, the 0.3 threshold). **c**, Worlds alive against pair-memory ring length P at L = 8, 10, 12 (Stage F; the native ring is 2L). **d**, Worlds alive at L = 9 (of 20) by ablation at 128 and 512 steps per encounter (Stage D).

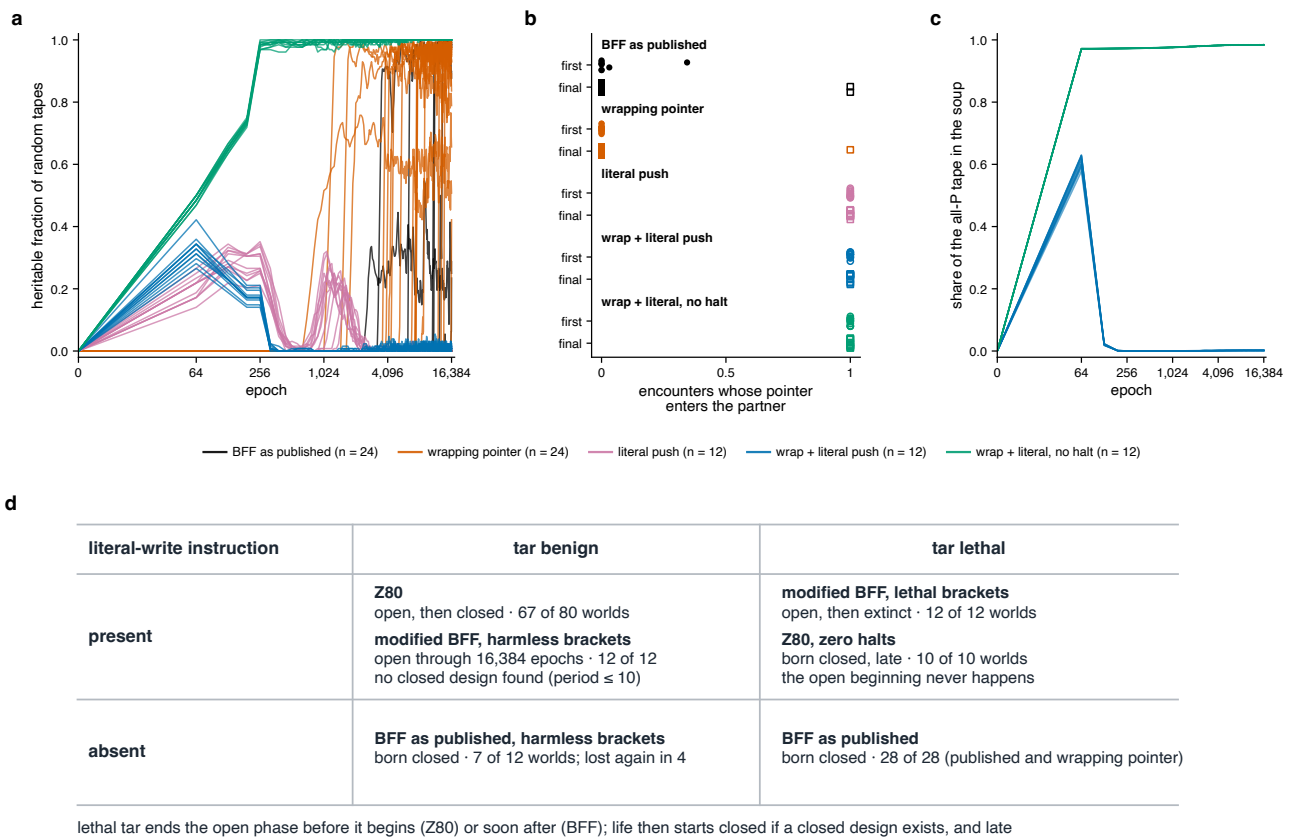


Fig. 5 | One instruction decides how life begins in BFF. **a**, Heritable fraction of 32 random tapes against epoch for every soup, by variant (as published, $n = 24$; wrapping pointer, $n = 24$; literal push, $n = 12$; wrap + literal push, $n = 12$; wrap + literal with unmatched brackets as no-ops, $n = 12$); the epoch axis is linear to 64 and logarithmic beyond, here and in **c**. **b**, Fraction of culture-test encounters in which the pointer enters the partner, for the first replicator (circles) and the final dominant (squares) of every life-producing soup, grouped by variant (open symbols, no loop instruction; filled, loop instruction). **c**, Share of the all-P class against epoch under wrap + literal (blue) and wrap + literal with no-op unmatched brackets (green), 12 worlds each (zero when the class is outside the ten largest): the same organism collapses when unmatched brackets halt the pointer and persists when they are no-ops. **d**, Map of the substrates by two properties, a literal-write instruction (present or absent) and the lethality of the tar to an open replicator, with the observed order of events and world counts in each cell; under lethal tar the Z80 (zero bytes halt execution) skips the open beginning and starts closed and late, whereas BFF's brief open beginning ends in extinction; without a literal channel and with harmless brackets, BFF is born closed in 7 of 12 soups and loses the replicator again in 4 of them.

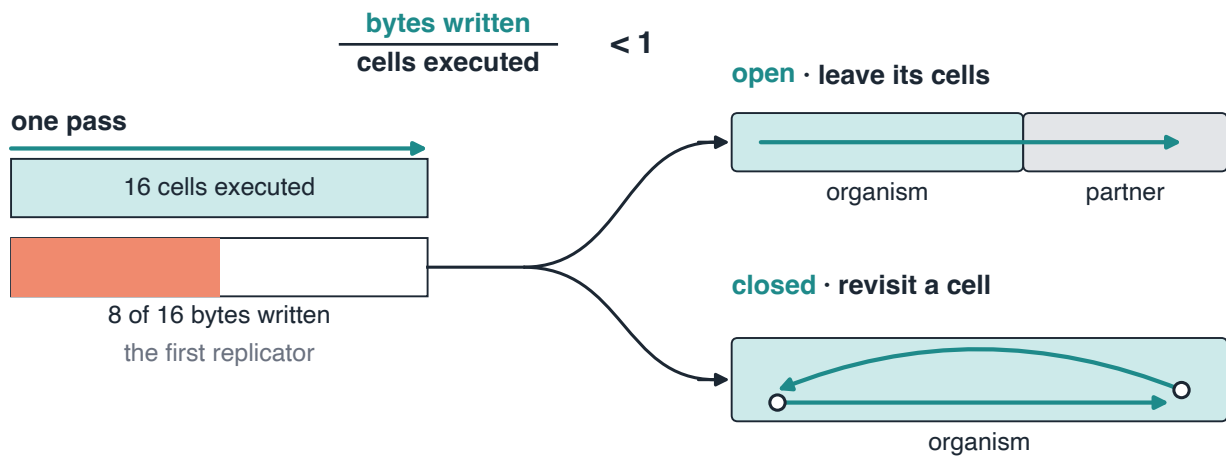


Fig. 6 | Closure requires a cycle. Theorem 2 as a diagram. A program that writes fewer bytes than the cells it executes cannot copy itself in one pass: the first replicator ($L = 16$) executes its 16 cells and has written 8 of the 16 bytes a copy needs (executor trace), so copying requires more execution, which can only continue outside its cells (open) or revisit one of them (closed, a cycle). Proofs, the closure-window model and the lemma on information inflow are in the Supplementary Information.

Extended Data

Extended Data Fig. 1 | Emergence by tape length and ablation. Kaplan–Meier fraction of worlds with a heritable replicator against step for every tape length (4–100) under the four Stage E control arms (nominal; constant mutation per byte; constant mutation per cell; constant budget per byte) and under the stack-writer ablation; 10 worlds per cell, 300,000 steps.

Extended Data Fig. 2 | The replicator zoo. First heritable replicator and final dominant tape of every Stage G world (80 worlds), as hex and disassembly, grouped by tape length, with copy success and self-damage from the partner test.

Extended Data Fig. 3 | Ring geometry. Worlds alive and median emergence step for every tested (tape length, ring length) pair of the ring sweep (52 cells) and Stage F; the pre-registered ring-arithmetic law (period divides the ring) against the observed law (period divides the pair length $2L$); the stack-pointer/instruction-pointer race that produces the dead zone.

Extended Data Fig. 4 | Budget and mutation at $L = 100$. Heritable and faithful worlds against steps per encounter (32–2,048) under the pusher and LDIR regimes; the sub-clonal regime at 64 steps (heritable fraction of random cells against dominant-tape share); median period of the first replicator and final high-order entropy against per-byte mutation rate (the error-threshold curve).

Extended Data Fig. 5 | Life detectors against the culture test. Area under the ROC curve for each recorded statistic (high-order entropy, non-unique tape fraction, bits per byte, species entropy, interaction-based statistics, dominant-tape share, quasispecies occupancy, motif share) against the heritable fraction of random cells, by tape length and by step; the inversion of high-order entropy under the zero flood.

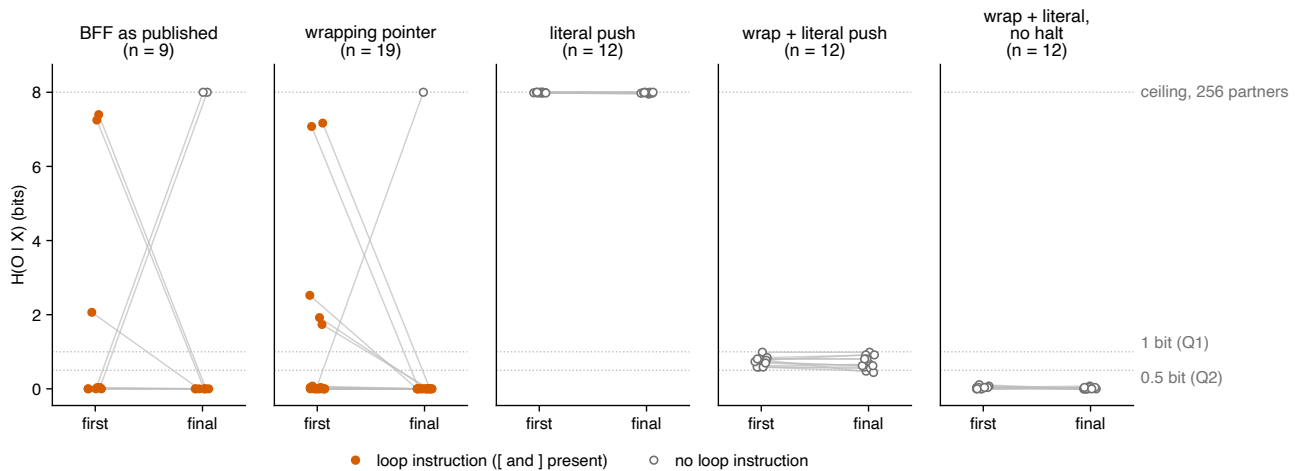
Extended Data Fig. 6 | Tar prevents assembly. Census of LDIR-bearing tapes in the ten most common tapes before emergence at $L = 9$, with and without stack writers (20 worlds per arm): first entry, episodes, destruction rate and zero bytes written per interaction.

Extended Data Fig. 7 | Invasion assays. The pusher and the LDIR unit seeded into 1% of cells at 32 and 128 steps per encounter and two mutation rates: occupancy against step, 20 worlds per cell.

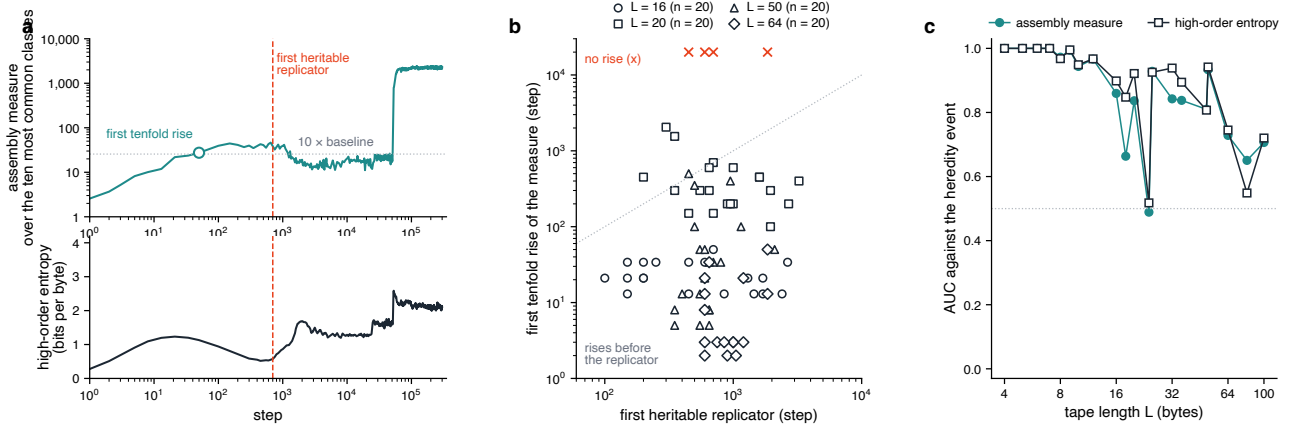
Extended Data Fig. 8 | Every two-byte word. Culture-test score and gen2 for all 65,536 two-byte words tiled to 16 bytes; the five fixed points against the zero partner; the 246 return-address smears.

Extended Data Fig. 9 | Searches in BFF. The 16.4-million-program straight-line search under a wrapping pointer (copy fraction and offspring byte diversity of the 130 programs passing the byte-match criterion) and the 1.3-million-tiling closed-replicator search under the benign-tar rule (open replicators by period; no closed heritable tiling).

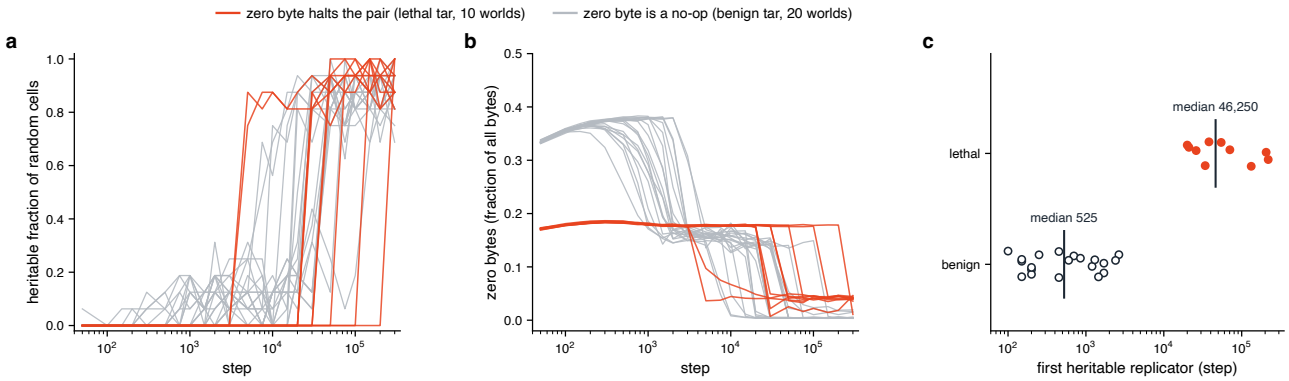
Extended Data Fig. 10 | Reproducibility and compute. Spread of emergence times across ten repeats of one seed against ten seeds in the Z80 soup (GPU scheduling nondeterminism); identity of epoch statistics between local Apple-GPU and Modal L40S runs of six BFF seeds over 16,384 epochs; GPU hours and cost per stage.



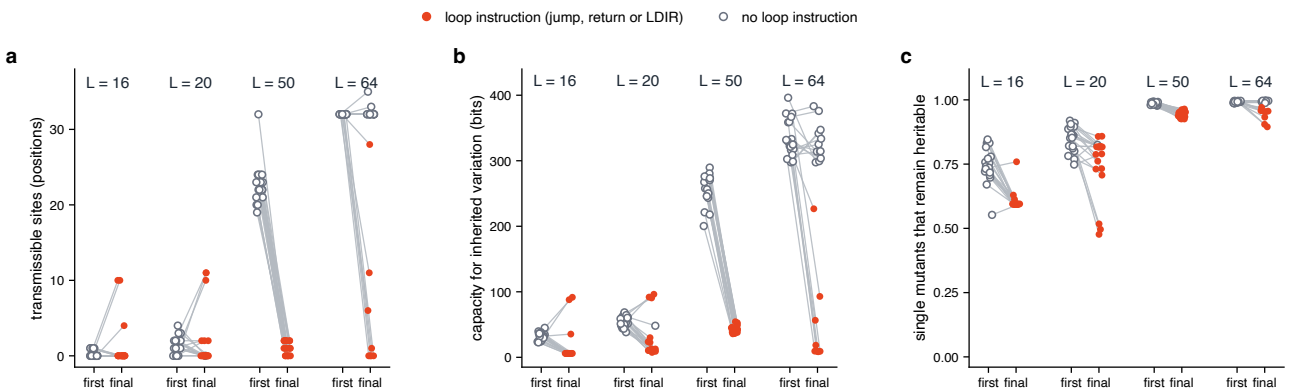
Extended Data Fig. 11 | Information inflow in the second machine. $H(o | x)$ over 256 random partners for the first replicator and the final dominant of every life-producing BFF soup, by variant (lines join the same soup; filled, both brackets present), with the 8-bit ceiling of 256 partners and the pre-registered 1-bit and 0.5-bit thresholds. The published variant is born with near-zero inflow; the literal-push variant without a wrapping pointer sits at the ceiling; the two wrapping literal variants flood the ring and have near-zero inflow although their pointer enters the partner in every encounter.



Extended Data Fig. 12 | A complexity-times-abundance measure against the culture test. **a**, The assembly measure over the ten most common classes (Re-Pair upper bound on the assembly index in Sharma et al.'s formula; top) and the high-order entropy (bottom) against step for the exemplar world ($L = 16$, seed 2001): the measure rises tenfold above its random-soup baseline at step 50 (circle), 650 steps before the first heritable replicator (dashed line). **b**, Step of the measure's first tenfold rise against the step of the first heritable replicator for all 80 Stage G worlds (dotted line, equality; crosses at the top edge, worlds without a rise). **c**, Area under the ROC curve of both statistics against the heredity event by tape length (Stage E, snapshot steps).



Extended Data Fig. 13 | Lethal tar in the first machine. Ten pre-registered worlds at $L = 16$ in which a zero byte fetched as an opcode halts the pair (Stage I, vermillion) against the twenty benign-tar worlds of Stage G at the same length (grey). **a**, Heritable fraction of 16 random cells against step. **b**, Zero-byte fraction against step. **c**, Step of the first heritable replicator in each world, with medians (46,250 against 525). Every lethal-tar first replicator is a block-copy replicator with a hardware loop, contains no zero byte, and copies 1.00 of partners with 0.00 self-damage from its first appearance.



Extended Data Fig. 14 | Closure trades variation for fidelity. Single-byte mutational scan of the first heritable replicator (first) and the final dominant (final) of every Stage G world, grouped by tape length (grey lines join the same world; filled, loop instruction; open, none). **a**, Transmissible sites: positions at which at least half of the tested alternative bytes leave the mutant heritable and are carried into at least half of its copies. **b**, Capacity for inherited variation, $\sum \log(1 + 255 \times \text{fraction of transmissible values})$ over positions, in bits. **c**, Robustness: fraction of single mutants that remain heritable (gen2 ≥ 0.3). All 255 alternative bytes per position at $L = 16$ and 20, 32 seeded values at $L = 50$ and 64; 32 partners per generation.

Extended Data Table 1 | Final-tape classes per tape length (Stage G). Control-flow and block-repeat content, partner-copy fraction and self-damage of the final dominant, number of worlds per class.

Extended Data Table 2 | Ablation sets. Opcode families removed in each ablation, with opcode counts and example mnemonics.

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Supplementary Information: theorems and proofs

What kind of statement each item is: a **Theorem** has a pen-and-paper proof and uses no simulation; a **Proposition** is an exact statement decided by finite computation, with the computation described; a **Model** is a result about a stylised model with stated assumptions, not about the simulator. Empirical premises are marked as such, and every number quoted is from the generated tables behind the main text.

Notation. An encounter executes a program held in a shared memory. The organism A occupies addresses 0 to $L - 1$ and the partner B the next L addresses; in the Z80 soups the $2L$ addresses form a ring, in BFF the 128 addresses do not wrap. Execution starts at address 0 with a fixed initial machine state (registers zero; the Z80 stack pointer at the last byte of B) and lasts at most T instruction executions ($T = 128$ in the Z80 soups, 2^{13} in BFF). A *control transfer* is a jump, call, return, repeat-in-place or halt; a *straight-line program* executes none, so that under default flow its pointer advances past each executed instruction. The *no-op context* is the partner all of whose bytes are no-ops (zero in the Z80, where byte 0 is NOP; zero in BFF, where every byte outside the ten instruction characters is a no-op). A tape x is a *self-writer* if its encounter with the no-op context ends with B holding a copy of x (up to a cyclic shift, as in the culture test), and a *full self-replicator in context e* if the encounter (x, e) ends with B holding a copy of x . It is *closed* if its pointer executes no address of B in any context and *open* if it does in some context. A value is *prescribed* if it is determined by x alone, the same for every partner.

Theorem 1 (no one-pass replicator in BFF). *Setting.* BFF as published: a 128-byte pair; the instruction pointer p and the two data heads h_0 and h_1 start at address 0; p advances by one after every instruction unless a bracket jumps; $< >$ and $\{ \}$ move h_0 or h_1 by one cell; $+ -$ change the byte under h_0 by one; $.$ copies the byte under h_0 into the cell under h_1 and $,$ the byte under h_1 into the cell under h_0 ; every other byte is a no-op; execution ends when p leaves the pair, on an unmatched bracket, or after 2^{13} steps. *Statement.* Let A be a 64-byte tape with at least two distinct byte values and B a partner that initially holds no byte of the copy in place. Any execution that leaves B holding a copy of A runs at least 129 instructions. Hence no straight-line execution, which runs at most 128 (one per address), produces a copy into such a partner, and a program that copies into every partner, as the culture test requires, is not straight-line; a constant tape produces no copy either. Writes, increments and moves are counted as disjoint sets of instructions below: w counts $.$ and $,$, p counts $+ -$, v counts the four moves. *Proof.* Let the copy have $D \geq 2$ distinct values. Each of the 64 cells of B must receive a write of a prescribed value. $+ -$ applied to a cell holding an unknown partner byte give an unknown byte, so a prescribed value reaches a cell only through $.$ or $,$, or through $+ -$ applied to a cell that already holds a prescribed value; therefore at least $w \geq 64$ write instructions run. A write lands on the cell under the writing head, so the 64 target cells are among the positions the heads occupy; both heads start at address 0 and each move changes one head's position by one, so after v

moves the heads have occupied at most $1 + v$ distinct positions. The first write of each of the D values must read it from a cell that holds it; before any cell of B holds a prescribed value only cells of A do, and a cell of A acquires a value it did not hold initially only by a write from a cell that held it or by an increment; so, if p increments run, the heads also occupy at least $D - p$ distinct cells of A that held distinct values initially. Target cells lie in B and these source cells in A , so $64 + D - p \leq 1 + v$. The instruction count is therefore at least $w + v + p \geq 64 + 63 + D \geq 129$. If $D = 1$ the tape is one repeated byte and, as a program, one repeated letter: a move letter never writes, a write letter with both heads at address 0 copies a cell onto itself, and $+$ or $-$ never write into B . ■ *Corollary 1.1.* Every self-replicator of BFF runs at least 129 instructions within 128 addresses, so it executes some address twice: it carries a loop. Whether the loop also keeps the pointer out of B is a separate, measured fact (0 of 28 first replicators were open).

Theorem 2 (closure requires a cycle). *Hypothesis (H1), default flow.* In the absence of an executed control transfer the pointer advances past each instruction, so that a straight-line program runs from the organism's region into the partner's. *Statement.* Under (H1), a closed full self-replicator of L bytes executes some address at least twice, provided either (i) a copy of it costs more than L instruction executions, or (ii) execution cannot end inside the organism's region before a budget $T > L$ is spent. *Proof.* A closed execution visits only the L addresses of A . If no address repeats, it comprises at most L instruction executions. Under (i) these cannot complete the copy; under (ii) the budget forces a further execution, which can only be at an address already executed. ■ *The machines.* BFF satisfies (i) by Theorem 1: a copy costs at least 129 executions, more than the whole pair has addresses. The Z80 soups satisfy (ii): the budget is 128 executions for tapes of at most 100 bytes, there is no exit instruction, and HALT holds the processor at its own address, which we count as a one-state cycle. *The bandwidth reading.* Condition (i) holds in any machine whose straight-line code writes fewer bytes than the addresses it consumes, since a pass over L addresses then writes fewer than L bytes. This is the form used in the main text, and it describes the replicators observed: the first replicator writes two bytes per four bytes of code (the executor trace of Fig. 6 shows eight bytes written when the pointer leaves its sixteen cells), BFF's copy instructions move one byte per instruction and need a head move per cell, and the literal push of modified BFF writes two bytes per three. It is not a machine-wide property of the Z80, in which a chain of pushes writes two bytes per byte of code; for the Z80 the theorem rests on (ii). *Corollary 2.1.* Under (H1) every straight-line program is open: with $T > L$ its pointer reaches the partner's region. A closed self-writer therefore contains an executed control transfer besides the instructions that write its content, and where a straight-line self-writer exists the closed one is longer by at least that instruction. For the Z80 the comparison is exact: by Proposition 3 the self-writers of period one or two are exactly five straight-line load-push words, so the smallest closed self-writer has period at least three (the closed replicators observed have period four or more).

Proposition 3 (the two-byte self-writers of the Z80; computer-assisted). Among the 65,536 two-byte words tiled to 16 bytes and executed for 128 instructions as A against the all-zero partner, exactly six leave B holding a tiling of the word: the trivial $00\ 00$, which B already holds, and the five load-push words $01\ c5, 11\ d5, 21\ e5$ ($LD\ rr, nn ; PUSH\ rr$) and $2a\ e5, e5\ 2a$ ($LD\ HL, (nn) ; PUSH\ HL$), decided by exhaustive execution of all 65,536 words. None contains a control transfer, so each runs on into the partner: every two-byte self-writer of this machine is open, and no tiling of period one or two is closed. The stochastic census agrees: the same five words copy into 0.66–0.81 of 256 random partners, none into 0.95 or more, and the 246 words that pass the heredity threshold by smearing return addresses produce no 75% copy.

Proposition 4 (open first: a count on the initial soup). In a soup of N tapes of L uniformly random bytes, the expected number of occurrences of a given k -byte word, at any of the $L - k + 1$ positions of a tape, is $N(L - k + 1) / 256^k$: for $N = 20,000$ and $L = 16$, 4.6 for a given two-byte word, 0.017 for a given three-byte word and 6.1×10^{-5} for a given four-byte word. A given two-byte word is present at step 0 with probability $1 - (1 - 256^{-2})^{N(L-1)} \approx 0.99$, a given three-byte word with probability ≈ 0.017 . *Consequence (conditional).* If the two-byte open self-writer has a positive growth rate in the soup (an empirical premise: it copies into 0.66–0.73 of random partners and invades from 1% of cells, Extended Data Fig. 7) and every closed self-writer has period at least three (Proposition 3), then an open self-writer is present from the start with probability ≈ 0.99 while a given closed design is absent with probability ≥ 0.98 (≥ 0.9999 at period four, the smallest observed), and the first replicator is open unless the open form fails to grow. This is the status of "open first": a theorem-grade count plus one empirical premise.

Model 5 (the closure window). *Assumptions.* An open replicator population of $n(t)$ tapes; each tape-step produces a closed functional variant with probability q (a mutation placing the right bytes at the right phase: q is of order $\text{sites} \times \mu \times 256^{-m}$ for an m -byte closing motif); a closed variant, once present, grows faster than the open form and fixes with probability of order one. *Statement.* The expected number of closed variants produced while the open population lasts is $q \int n\ dt$, and closure occurs with probability $\approx 1 - \exp(-q \int n\ dt)$; the integral is the window. If the open form's own copies are viable in the soup it creates, n stays near the carrying capacity, the window grows without bound and closure is a matter of waiting; if the soup becomes lethal to the open form before $q \int n\ dt$ reaches order one, the open form dies without closing. *What the experiments found.* Z80, benign zeros: closure in 20 of 20 worlds by 300,000 steps at $L = 16$ and 50, in 19 of 20 by one million at $L = 20$ and in 8 of 20 at $L = 64$, where the open form is least damaged and the motif

must land in the longest tape (smallest q). BFF with the literal push and lethal brackets: n fell from $0.6 N$ at epoch 64 to $3 \times 10^{-4} N$ by epoch 256, a window of order 10^7 tape-epochs, and 0 of 12 closures. BFF with benign brackets: the window is unbounded, yet no closure by control flow in 12 of 12, consistent with $q = 0$, since no closed heritable tiling exists in that language up to period 10 (Methods); the open form persists for 16,384 epochs. Z80 with lethal zeros (Stage I): the open form never establishes, so the window is empty from the start, and the first replicator is closed and late (median 46,250 steps against 525; 10 of 10 worlds). The model does not predict which regime a machine is in; that depends on how the open replicator's own by-products act on it, which is what the experiments measure.

Theorem 6 (architecture decides the first replicator). *Hypotheses.* (H1) Default flow, as in Theorem 2, with execution ending when the pointer leaves the pair (as in BFF) or continuing around a ring (as in the Z80 soups). (H2) *Copy cost*: writing a copy of L bytes with at least two distinct values costs more instruction executions than the pair has addresses, $2L$ (true of BFF by the proof of Theorem 1, and of any machine whose write instructions place one byte at a head that moves one cell per instruction). (H3) *Self-consistent literal write*: an instruction that writes its own operand bytes verbatim at a write pointer that starts in, or advances into, the partner's region, and whose operand can be its own encoding in the phase executed (`LD BC, nn ; PUSH BC` with $nn = c5\ 01$; `P a b` with $a = b = P$). *Statement.* (i) Under (H1) and either proviso of Theorem 2, every closed self-replicator contains a cycle. (ii) Under (H1) and (H3), the tiling of the literal instruction is a straight-line program, hence open, that writes its own content at the advancing write pointer; it is a self-writer wherever the encounter lets the written bytes cover the partner's region, which holds in the Z80 (eight pushes over two passes of the $2L$ -byte ring; Proposition 3) and in BFF with P under both pointer rules (12 of 12 soups in each; the straight-line search, Methods). (iii) Under (H1) and (H2) without (H3), in a machine whose execution ends when the pointer leaves the pair, no straight-line program is a self-writer and every self-replicator contains a cycle: born loop-bearing (whether its pointer also stays out of the partner is a separate, measured property). (iv) Under uniform random initialisation over an alphabet of size σ , the expected number of occurrences of a fixed k -symbol pattern among N regions of length L is $N(L - k + 1) \sigma^{-k}$, so a minimal open pattern shorter than every closed one is present with probability higher by a factor exponential in the length difference. *Proof.* (i) is Theorem 2. (ii) The tiling executes no control transfer, so by (H1) its pointer runs into the partner's region: it is open; its writes are its operands, by (H3) the tiling itself; completion of the copy is the per-machine fact cited. (iii) In the no-op context a straight-line program runs at most $2L$ instructions before the pointer leaves the pair, and by (H2) these cannot complete a copy; so a self-writer executes a control transfer, and since a copy costs more than $2L$ executions within $2L$ addresses, some address repeats. (iv) is linearity of expectation. ■ *What it decides.* Whether a substrate's first replicator can be open is decided by (H3). The Z80 has a self-consistent literal write (open first, 80 of 80 worlds); BFF has none, since its instructions increment in place or copy from memory (28 of 28 first replicators loop-bearing, and 0 of 28 enter the partner by measurement); BFF with one literal-write instruction added is open first (12 of 12). The theorem says nothing about how long the open phase lasts (Model 5). *Scope.* (H1) is a property of the von Neumann architecture, shared code and data under sequential control, and holds in Core War, Tierra, Avida, the Z80 and BFF alike. (H2) is a property of the copy instructions and must be checked per machine: it holds in BFF by Theorem 1 and fails in the ring Z80, whose two-byte self-writer completes its copy on a second pass of the ring, so for the Z80 the open-first conclusion rests on Propositions 3 and 4 and on observation. For a chemistry the hypotheses are a reading, not a theorem: (H1) would be a copier that runs off its template into adjacent material by default, (H3) a template whose content specifies its own complement without an interpreter.

Lemma 7 (closure as zero information inflow). Let the offspring of x in context e be $o = F(x, e)$, with e drawn from a context distribution, and define the inflow $\lambda(x) = I(o; e | x)$, the information the offspring carries about the context given the parent. (i) If $\lambda(x) > 0$ then $P(o = x | x) < 1$: a replicator with inflow cannot be perfectly faithful across contexts. (ii) If $P(o = x | x) = 1$ then $\lambda(x) = 0$: perfect fidelity across contexts implies zero inflow. (iii) Fano's inequality bounds the fidelity: $P(o \neq x | x) \geq (H(o | x) - 1) / \log|O|$, where O is the set of possible offspring. *Proof.* If $o = x$ with probability one given x , then o is a function of x alone and $I(o; e | x) = 0$, which is (ii); (i) is its contrapositive; (iii) is Fano's inequality with $H(o | x) \geq I(o; e | x)$. ■ *Measurement.* For a deterministic executor $H(o | x, e) = 0$, so $\lambda(x) = H(o | x)$, the entropy of the offspring over partners; the partner test estimates it as the plug-in entropy over 256 uniformly random partners (ceiling 8 bits). *Reading, with the measured qualification.* The lemma is Krakauer et al.'s individuality, the organism's future determined by its own past rather than the environment's, specialised to one generation and made measurable by intervention. Pointer closure (execution never enters the partner) is neither necessary nor sufficient for $\lambda = 0$: it is not sufficient because offspring positions the organism never writes keep the partner's bytes (7 of 28 pointer-closed BFF first replicators have $H(o | x) > 0$, three of them ≈ 7 bits from a single unwritten byte), and it is not necessary because a copier that overwrites everything it touches has $\lambda \approx 0$ although its pointer enters the partner (the wrapping literal pusher of BFF, 0.04 bits). Inflow comes from partial copying; openness of control flow is the mechanism that makes the Z80 first replicator's copies partial. Measured: the first replicator has $\lambda > 0$ in 80 of 80 worlds (median 7.0 bits) and the loop-bearing successor $\lambda = 0$ in 63 of 67 (one offspring string in all 256 contexts), the four exceptions copying all but one to three bytes (Fig. 3c).